KAIZEN MATHS

Lesson starter

Basic Integration and Definite Integrals

Indefinite Integrals

Name _____

S1

Find $\int (-6x^3 - 4x^2)\,dx$.

S2

Find $\int (-6x^{\frac{4}{3}} + 3x^{\frac{4}{3}})\,dx$.

S3

Find $\int (3x^{-4} + 2x^{-1})\,dx$.

S4

Find $\int (5x^5 + 4x^2)\,dx$.

Show your working

KAIZEN MATHS

Lesson starter

Basic Integration and Definite Integrals

Indefinite Integrals

Name _____

S1

Find $\int (-6x^3 - 4x^2)\,dx$.

S2

Find $\int (-6x^{\frac{4}{3}} + 3x^{\frac{4}{3}})\,dx$.

S3

Find $\int (3x^{-4} + 2x^{-1})\,dx$.

S4

Find $\int (5x^5 + 4x^2)\,dx$.

ALG-06EF-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Basic Integration and Definite Integrals

Indefinite Integrals

Name _____

E1

Find $\int (3x^{\frac{4}{3}} + 2x^{\frac{5}{2}})\,dx.$

E2

Find $\int (-4x^{-1} + 6x^{-2})\,dx.$

E3

Find $\int (-5x^4 - 5x^2 - 6x)\,dx.$

KAIZEN MATHS

Exit ticket

Basic Integration and Definite Integrals

Indefinite Integrals

Name _____

E1

Find $\int (3x^{\frac{4}{3}} + 2x^{\frac{5}{2}})\,dx.$

E2

Find $\int (-4x^{-1} + 6x^{-2})\,dx.$

E3

Find $\int (-5x^4 - 5x^2 - 6x)\,dx.$

ALG-06EF-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Basic Integration and Definite Integrals

Indefinite Integrals

Name _____

E1

Find $\int (3x^{\frac{4}{3}} + 2x^{\frac{5}{2}})\,dx.$

E2

Find $\int (-4x^{-1} + 6x^{-2})\,dx.$

E3

Find $\int (-5x^4 - 5x^2 - 6x)\,dx.$

ALG-06EF-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Basic Integration and Definite Integrals

Indefinite Integrals

Name _____

E1

Find $\int (3x^{\frac{4}{3}} + 2x^{\frac{5}{2}})\,dx.$

E2

Find $\int (-4x^{-1} + 6x^{-2})\,dx.$

E3

Find $\int (-5x^4 - 5x^2 - 6x)\,dx.$

ALG-06EF-L1 · E

Show your working

Teacher answers & working

Basic Integration and Definite Integrals · Indefinite Integrals

ALG-06EF-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Find $\int (-6x^3 - 4x^2)\,dx$.

Answer:

$= -\frac{3}{2}x^4 - \frac{4}{3}x^3 + C$

Working:

Step 1: Use the reverse power rule on each term: add 1 to the power and divide by the new power.

Use the reverse power rule. Add 1 to the power, then divide by the new power.

For the term $-6x^3$, add 1 to the power.

Substitute into the reverse power rule.

Simplify the coefficient: $\frac{-6}{4}x^4$

Write the simplified term.

For the term $-4x^2$, add 1 to the power.

Substitute into the reverse power rule.

Write the simplified term.

Combine the integrated terms.

Add the constant of integration $+C$.

Final answer: $= -\frac{3}{2}x^4 - \frac{4}{3}x^3 + C$

S2

Find $\int (-6x^{\frac{4}{3}} + 3x^{\frac{4}{3}})\,dx$.

Answer:

$= \frac{-18}{7}x^{\frac{7}{3}} + \frac{9}{7}x^{\frac{7}{3}} + C$

Working:

Step 1: Treat fractional powers exactly like powers: add 1 to the exponent, then divide by the new exponent.

Use the reverse power rule. Add 1 to the power, then divide by the new power.

For $-6x^{\frac{4}{3}}$, add 1 to the fractional power.

Divide by the new power, which is the same as multiplying by its reciprocal.

Write the simplified term.

For $3x^{\frac{4}{3}}$, add 1 to the fractional power.

Divide by the new power, which is the same as multiplying by its reciprocal.

Write the simplified term.

Final answer: $= \frac{-18}{7}x^{\frac{7}{3}} + \frac{9}{7}x^{\frac{7}{3}} + C$

Teacher answers & working

Basic Integration and Definite Integrals · Indefinite Integrals

ALG-06EF-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Find $\int (3x^{-4} + 2x^{-1})\,dx$.

Answer:

$= -\frac{1}{x^3} + 2\ln|x| + C$

Working:

Step 1: Use the reverse power rule for powers other than x^{-1}; remember that x^{-1} integrates to $\ln|x|$.

Use the reverse power rule. Add 1 to the power, then divide by the new power.

Use the logarithm rule only when the power is -1: $\int ax^{-1}\,dx = a\ln|x|$.

For $3x^{-4}$, add 1 to the power.

Substitute into the reverse power rule.

This term has power -1, so use the logarithm rule.

Final answer: $= -\frac{1}{x^3} + 2\ln|x| + C$

S4

Find $\int (5x^5 + 4x^2)\,dx$.

Answer:

$= \frac{5}{6}x^6 + \frac{4}{3}x^3 + C$

Working:

Step 1: Use the reverse power rule on each term: add 1 to the power and divide by the new power.

Use the reverse power rule. Add 1 to the power, then divide by the new power.

For the term $5x^5$, add 1 to the power.

Substitute into the reverse power rule.

Write the simplified term.

For the term $4x^2$, add 1 to the power.

Substitute into the reverse power rule.

Write the simplified term.

Combine the integrated terms.

Add the constant of integration $+C$.

Final answer: $= \frac{5}{6}x^6 + \frac{4}{3}x^3 + C$

Teacher answers & working

Basic Integration and Definite Integrals · Indefinite Integrals

ALG-06EF-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E1

Find $\int (3x^{\frac{4}{3}} + 2x^{\frac{5}{2}})\,dx$.

Answer:

$= \frac{9}{7}x^{\frac{7}{3}} + \frac{4}{7}x^{\frac{7}{2}} + C$

Working:

Step 1: Treat fractional powers exactly like powers: add 1 to the exponent, then divide by the new exponent.

Use the reverse power rule. Add 1 to the power, then divide by the new power.

For $3x^{\frac{4}{3}}$, add 1 to the fractional power.

Divide by the new power, which is the same as multiplying by its reciprocal.

Write the simplified term.

For $2x^{\frac{5}{2}}$, add 1 to the fractional power.

Divide by the new power, which is the same as multiplying by its reciprocal.

Write the simplified term.

Final answer: $= \frac{9}{7}x^{\frac{7}{3}} + \frac{4}{7}x^{\frac{7}{2}} + C$

E2

Find $\int (-4x^{-1} + 6x^{-2})\,dx$.

Answer:

$= -4\ln|x| - \frac{6}{1\,x^{1}} + C$

Working:

Step 1: Use the reverse power rule for powers other than x^{-1}; remember that x^{-1} integrates to $\ln|x|$.

Use the reverse power rule. Add 1 to the power, then divide by the new power.

Use the logarithm rule only when the power is -1: $\int ax^{-1}\,dx = a\ln|x|$.

This term has power -1, so use the logarithm rule.

For $6x^{-2}$, add 1 to the power.

Substitute into the reverse power rule.

Final answer: $= -4\ln|x| - \frac{6}{1\,x^{1}} + C$

Teacher answers & working

Basic Integration and Definite Integrals · Indefinite Integrals

ALG-06EF-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

Find $\int(-5x^4 - 5x^2 - 6x)\,dx$.

Answer:

$= -x^5 - \frac{5}{3}x^3 - 3x^2 + C$

Working:

Step 1: Use the reverse power rule on each term: add 1 to the power and divide by the new power.

Use the reverse power rule. Add 1 to the power, then divide by the new power.

For the term $-5x^4$, add 1 to the power.

Substitute into the reverse power rule.

Simplify the coefficient: $\frac{-5}{5}x^5$

Write the simplified term.

For the term $-5x^2$, add 1 to the power.

Substitute into the reverse power rule.

Write the simplified term.

For the term $-6x$, add 1 to the power.

Substitute into the reverse power rule.

Simplify the coefficient: $\frac{-6}{2}x^2$

Write the simplified term.

Combine the integrated terms.

Add the constant of integration $+C$.

Final answer: $= -x^5 - \frac{5}{3}x^3 - 3x^2 + C$